

Homework Assignment – Electromagnetic Spectrum

Homework 5 (Due Thursday, Week 8):

1. Arrange these waves in order of increasing frequency: X-rays, Radio waves, Infrared, Ultraviolet.
2. A microwave oven uses waves with wavelength 0.12 m . Find the frequency.
3. The human eye is most sensitive to light of wavelength $5.5 \times 10^{-7} \text{ m}$. What is the frequency of this light?
4. Explain why gamma rays are more dangerous than radio waves.
5. Which part of the spectrum is used in night vision goggles? Why?